they photographed the year before … and the year before that. If they got it right last time, you might well ask, why are they doing it again? The answer is: The sky changes. In any given year there might be worldlets wholly unknown, never seen before, that approach the Earth and are spied by these dedicated observers.

On March 25, 1993, a group of asteroid and comet hunters, looking at the photographic harvest from an intermittently cloudy night at Mount Palomar in California, discovered a faint elongated smudge on their films. It was near a very bright object in the sky, the planet Jupiter. Carolyn and Eugene Shoemaker and David Levy then asked other observers to take a look. The smudge turned out to be something astonishing: some twenty small, bright objects orbiting Jupiter, one behind the other, like pearls on a string. Collectively they are called Comet Shoemaker-Levy 9 (this is the ninth time that these collaborators have together discovered a periodic comet).

But calling these objects *a* comet is confusing. There was a horde of them, probably the fragmented remains of a single, hitherto undiscovered, comet. It silently orbited the Sun for 4 billion years before passing too close to Jupiter and being captured, a few decades ago, by the gravity of the Solar System's largest planet. On July 7, 1992, it was torn apart by Jupiter's gravitational tides.

You can recognize that the inner part of such a comet would be pulled toward Jupiter a little more strongly than the outer part, because the inner part is closer to Jupiter than the outer part. The difference in pull is certainly small. Our feet are a little closer to the center of the Earth than our heads, but we are not in consequence torn to pieces by the Earth's gravity. For such tidal disruption to have occurred, the original comet must have been held together very weakly. Before fragmentation, it was, we think, a loosely consolidated mass of ice, rock, and organic matter, maybe 10 kilometers (about 6 miles) across.

The orbit of this disrupted comet was then determined to high precision. Between July 16 and 22, 1994, all the cometary fragments, one after another, collided with Jupiter. The biggest